

NABEEL SHAN

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SUMMARY

Aspiring AI/ML Engineer with a strong foundation in deep learning, computer vision, and natural language processing. Passionate about building intelligent systems that solve real-world problems and drive innovation. Skilled in Python, TensorFlow, PyTorch, and modern ML workflows, with hands-on experience developing custom neural networks, training pipelines, and end-to-end AI applications. Eager to contribute to impactful research, collaborate on open-source projects, and pursue advanced studies through a fully funded Master's in Artificial Intelligence. Known for a growth mindset, clean code practices, and a relentless drive to learn, build, and lead in the field of AI.

EDUCATION

National University of Sciences and Technology (NUST), Military College of Signals

2027

Bachelor, Software Engineering

- **GPA:** 3.52
- **Achievements:** Dean's List: Fall 2023, Spring 2027
- **Coursework:** DSA, Database Systems, Software Engineering, Object-Oriented Programming, Operating Systems, Linear Algebra, Probability And Statistics, Computer Networks

TECHNICAL SKILLS

- **Languages/Frameworks/Libraries:** Python, C++, Java, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Matplotlib
- **AI/ML:** CNNs, RNNs, ResNets, YOLO, GRUS, LSTMs, Attention Mechanisms, Seq2Seq Models, Word Embeddings (GloVe, Word2Vec), Tokenization, Transfer Learning, Backpropagation, Fine-Tuning, Model Evaluation, Regularization, Hyperparameter Tuning, Data Visualization, Data Analysis, Machine Learning, Deep Learning, LLM APIs
- **Tools:** Google Colab, Jupyter, Git, GitHub, VS Code, Application Deployment, Cloud Services.
- **Soft Skills:** Strong Communications Skills, problem-solving, Adaptability, Team collaboration, Growth-oriented learner

PROJECTS

YOLO Object Detection Pipeline (TensorFlow, Keras)

- Engineered an end-to-end YOLOv2 object detection pipeline from scratch, covering image preprocessing, model inference, and post-processing with Non-Max Suppression.
- Implemented a custom yolo_head decoder and bounding box visualization.
- Generated side-by-side comparison images and videos for prediction evaluation and presentation.

ResNet-50 from Scratch - CIFAR-10 (3 classes)

- Implemented ResNet-50 architecture manually.
- Trained on a filtered subset of CIFAR-10 (classes 0, 1, 2).
- Created custom identity/convolutional blocks and visualized training metrics.
- Integrated He initialization, batch normalization, and shortcut connections.

Transfer Learning with MobileNetV2 - Alpaca Classifier

- Applied transfer learning and fine-tuning on MobileNetV2 (ImageNet weights) for binary classification of Alpaca vs Not-Alpaca images.
- Built a custom head, used data augmentation, and selectively unfroze layers for better convergence.
- Tracked performance across phases with visualizations of loss, accuracy, and validation metrics.

Face Recognition using CNNs + Transfer Learning

- Built a complete face recognition system using a pretrained FaceNet model.
- Generated 128D face embeddings and compared them via L2 (Euclidean) distance for identity verification and recognition.
- Enabled real-time matching of unknown faces against a known database using threshold-based decision logic.
- Ensured robustness to lighting and angle variations.

Emojiify NLP – Emoji Prediction from Text (NumPy, TensorFlow, Keras)

- Developed two deep learning models to predict emojis from short English sentences, simulating emotion-aware text understanding.

- **V1:** Implemented baseline model using averaged GloVe word embeddings + softmax classifier in pure NumPy (82.1% test accuracy).
- **V2:** Built an LSTM-based sequence model with pretrained GloVe embeddings and Keras layers, achieving **85.7% test accuracy**.
- Visualized training metrics, interpreted softmax outputs, and demonstrated sentence-to-emoji mapping with real examples.

Attention-Based Date Format Translator (TensorFlow, Keras)

- Built a custom sequence-to-sequence model with attention to translate natural language date formats (e.g., "21st of August 2016") into machine-readable formats ("2016-08-21").
- Designed a Bi-LSTM encoder, custom attention layer, and LSTM decoder with softmax output for character-level sequence generation.
- Trained on synthetically generated dates using faker and babel libraries with multi-format input and standardized YYYY-MM-DD output.
- Visualized attention weights and verified predictions via sample inference for robust date normalization.

Debiasing GloVe Word Embeddings (Bias Mitigation in NLP)

- Reproduced and implemented core techniques from the seminal paper "*Man is to Computer Programmer as Woman is to Homemaker?*" (Bolukbasi et al., 2016) to **detect and reduce gender bias** in GloVe word embeddings.
- Built custom analogy solvers to expose bias, then applied **neutralization** to debias gender-neutral terms and **equalization** to symmetrically align gendered word pairs (e.g., king/queen, he/she).
- Visualized cosine similarities and embedding shifts before and after debiasing with intuitive bar plots and vector projections.
- Gained hands-on experience in **ethical AI**, bias detection, and fairness techniques within NLP systems - demonstrating a commitment to responsible AI development.

CARLA Road Segmentation with Custom U-Net

- Designed and implemented a complete U-Net architecture from scratch for pixel-level road segmentation using CARLA simulator data.
- Manually built encoder-decoder blocks with skip connections, transposed convolutions, and dropout layers.
- Achieved accurate road and lane boundary detection on synthetic driving scenes with custom training, evaluation, and visualization pipeline

Deep Neural Network from Scratch (NumPy)

- Engineered a complete deep neural network framework from scratch using NumPy.
- Implemented forward and backward propagation, dropout regularization, L2 penalty, and mini-batch training.
- Integrated multiple optimizers including Gradient Descent, Momentum, RMSprop, and Adam.
- Trained and evaluated on a binary cats dataset with visual comparisons of loss curves and decision boundaries.

CNN from Scratch – Sign Language Digit Classifier (NumPy)

- Engineered a fully functional **Convolutional Neural Network (CNN) from scratch** using only NumPy to classify **RGB images of hand signs (0–5)** in sign language.
- Implemented the **entire deep learning pipeline manually** - including Conv2D, ReLU, Max Pooling, Dense, Softmax, and full **backpropagation** using chain rule-based gradient calculation.
- Visualized **training curves**, **forward/backward shape flows**, and **feature extraction logic** to gain a first-principles understanding of CNNs.
- Achieved smooth convergence and solid classification performance - showing mastery over low-level neural network mechanism and a unique showcase of bottom-up AI thinking, mathematical understanding, and raw implementation skills.

Vanilla RNN from Scratch – Character-Level Text Generator (NumPy)

- Designed and implemented a complete **Vanilla Recurrent Neural Network (RNN)** from scratch using NumPy - no TensorFlow, no PyTorch.
- Built full support for **character-level language modeling**, including forward pass, **backpropagation through time (BPTT)**, softmax-based prediction, and temperature-controlled text sampling.
- Trained on custom text sequences and generated coherent outputs, demonstrating learned dependencies across character streams.
- Visualized training loss over epochs and explored generative outputs conditioned on different starting tokens.
- Showcased a solid grasp of low-level sequence modeling, RNN internals, and the mathematical mechanics of training recurrent architecture.

CERTIFICATIONS

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- **Machine Learning Specialization:** Stanford University via Coursera (Jan 2025)
 - **IBM Machine Learning Specialization:** IBM via Coursera (May 2025)
 - **Deep Learning Specialization:** DeepLearning.AI via Coursera (Jun 2025)
 - **Harvard CS50 – Introduction to Computer Science:** Harvard University via edx (2024)
 - **Data Analysis with Python:** IBM via Coursera (Jun 2025)
 - **Free Python Course with Certificate – Updated – GeeksforGeeks**

Also a football & badminton tournament winner, and passionate reader of tech & psychology non-fiction.